

Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2021 - Present	B.Tech	Indian Institute of Technology Kanpur	9.3/10
2021	CBSE (XII)	Sidhartha Public School Delhi	89.8%
2019	CBSE (X)	Model School Rohtak, Haryana	95.8%

Scholastic Achievements

- Stood **3rd** in **Tower Research Scholarship Test**(2022) among 129 IIT Kanpur CSE dept students, winning **₹1 lakh** as a prize
- Received the **Academic Excellence Award** by IIT Kanpur for outstanding and exceptional academic performance in 2021-22
- Secured an **All India Rank 771** in the **Joint Entrance Examination Advanced 2021** amongst 2,50,000+ students qualified
- Secured an **All India Rank 219** in the **Joint Entrance Examination Main 2021** amongst 12,00,000+ students nationwide
- Have achieved maximum **Codeforces** rating of **1542+** with ranks **866**(CF-875 Div2) and **934**(CF-882 Div2) on *geetika0411*
- Shortlisted among top 75 students chosen from a massive pool of 2500+ applicants for the UberShe++ Coding Challenge 2023
- Received **KVPY**(Kishore Vaigyanik Protsahan Yojana) Fellowship 2021 from **IISc Bangalore**, with an **All India Rank 863**

Key Projects

- Course Allocation

Research Project, Prof. Samir Khuller, Northwestern University, Chicago, USA

 (May'23-Ongoing)
- Studied clash-free course allocation algorithms for once-a-week classes, taking into account various credit caps(uniform, arbitrary, infinite) and utility(uniform, binary) models(ranging from polynomial time solvable to np-hard) which guarantee envy freeness
 - Analysed best-known algorithms for the **t-interval graphs** to enhance course allocation algorithms for multiple classes per week
 - Modeled relaxed **linear programming** algorithm(2t approx) for Max Weight Independent Set in t-interval graph using **gurobipy**
- What's Next

Course Project, Prof. Indranil Saha, Dept. of CSE IIT Kanpur

 (Feb'23-Apr'23)
- Developed a **web application** using the **MERN Stack** which enables event scheduling and delivers personalised notifications
 - Implemented integrative user interface using **React.js** which has like/dislike functionality, event details page and admin dashboard
 - Integrated **MongoDB** as the robust **backend database** to store event details, facilitating seamless retrieval, manipulation, and storage of data according to various parameters while leveraging **Express.js** and **Node.js** to build a secure **server-side API**
- Model Mania

Course Project, Prof. Purushottam Kar, Dept. of CSE IIT Kanpur

 (Feb'23-Apr'23)
- Examined efficacy of **LinearSVC** and **LogisticRegression** models by optimizing hyperparameters like loss, tolerance, penalty
 - Trained **Decision tree** model which plays word guessing game(99% accurate), by maximizing information gain using Gini index
 - Achieved **1.896** MAE with a training time of **1.23s** for KNN by testing various models (**K-Nearest Neighbors**, **SVM**, **Kernel** and **DNN**) on a dataset with features (timestamp, temperature, humidity, sensor readings etc) for predicting O₃ and NO₂ levels
- Bubble Machine

Course Project, Prof. Urbi Chatterjee, Dept. of CSE IIT Kanpur

 (Feb'23-Apr'23)
- Developed **Verilog HDL** processor following custom Instruction Set Architecture, having word and instruction size of **32 bits**
 - Designed **FSM** to **fetch** instruction from instruction memory and subsequently **decode** and **execute** it using **MIPS32 ALU**
 - Tested and validated the processor on **Bubble sort** code written in MIPS by storing the resulting data back into data memory
- Web 3.0

Mentored by Science and Technology Council, IIT Kanpur

 (Jun'22-Jul'22)
- Learnt about the concepts and implementations of **Blockchain**, **Consensus mechanisms**(PoW and PoS), **Ethereum** blockchain
 - Implemented a **Loan contract** using **Solidity** which is derived from **Meta Coin** contract, and capable of accepting loan requests, displaying the owner's balance, allowing the users to view the owed amounts, and enabling settlement of dues with the creditors

Technical Skills

- **Programming Languages and Libraries:** C, C++, Python, NumPy, Pandas, Matplotlib, Scikit-learn, Easyocr, Gurobipy
- **Development and Utilities:** Git, Bash, HTML, CSS, JavaScript, Nodejs, Expressjs, MongoDB, Verilog, MIPS, Solidity, $\text{\LaTeX}$

Relevant Courses

A*-EXCEPTIONAL PERFORMANCE

Data Structure and Algorithms(A*) Introduction to Machine Learning Fundamentals of Computing	Probability for Computer Science Software Development and Operations Linear Algebra and Differential Equations	Discrete Mathematics Computer Organisation Introduction to Economics
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Positions of Responsibility

- Secretary, Association of Computing Activities, IIT Kanpur

(Feb'23-Present)
- Mentored group of 15 students for competitive programming project, organized departmental farewell, and anchored happy hours
- Secretary, Programming Club, IIT Kanpur

(2022-23)
- Organised hackathons, lectures, competitions, and hands-on workshops for the programming enthusiasts in the campus community
 - Proactively contributed and maintained **open source projects** used by the campus community of approximately **6000** people
- Web Secretary, Entrepreneurship Cell, IIT Kanpur

(2022-23)
- Responsible for managing all E-Cell **websites** and **portals** with the monthly views on the main website being **10,000+** users
 - Managing and planning cell's major events like **ESummit**, **Upstart**, Campus Hangouts etc. which witness over **5000** participants